LEARNER GUIDE

Forming an equation

Equations, Inequalities and Simultaneous Relationships · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

An equation is a question in disguise. When a situation asks 'what number makes this work?', you can capture it exactly: **name the unknown** (let x be the thing you don't know), then **write what the situation says** as an equation. 'Multiply by 3' becomes the 3 in front of x; 'add 4' becomes + 4; 'the result is 19' becomes = 19. Once written, you solve it as usual — and then you **interpret**: read the answer back into the situation, because the question sometimes asks for something other than x (an age, a length, the larger number), and always **check** it makes the story true.

What you are learning to notice

- Naming the unknown ('let x be…') is the first and most useful move.
- The words map straight onto the equation: 'multiply by 3' $\rightarrow 3x$; 'add 4' $\rightarrow$ + 4; 'is 19' $\rightarrow$ = 19.
- Solving finds x; **interpreting** turns x into the answer the situation actually asked for.
- A good answer is **checked against the story**, not just the algebra.

Moving through it, one step at a time

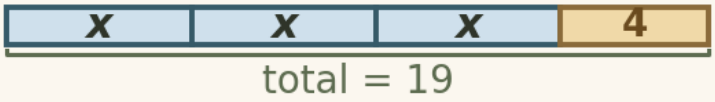
First name the unknown and write the equation, one phrase at a time. Solve it using the methods you already know — clear any fraction or bracket, collect, then undo step by step. Then read the question again: does it want x, or something built from x? Give that, and check your answer makes the original situation true.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Think of a number

'I think of a number, multiply it by 3, then add 4, and the result is 19.' The bar shows 3 x-blocks and a 4-block making 19.



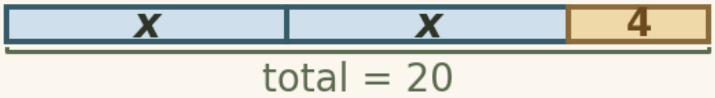
total = 19

1. Let the number be x, and write the situation.	$3x + 4 = 19$
2. Subtract 4 from both sides.	$3x = 15$
3. Divide both sides by 3.	$x = 5$
4. Check against the story.	$5 \times 3 = 15$, then $+ 4 = 19$ ✓

$x = 5$ — form $3x + 4 = 19$, then solve; $5 \times 3 + 4 = 19$ matches the story.

Example 2 — Interpreting the answer

'Emma is x; her sister is 4 years older; together they are 20.' The bar shows two x-blocks and a 4-block making 20.



total = 20

1. Write the total.	$x + (x + 4) = 20$, so $2x + 4 = 20$
2. Subtract 4 from both sides.	$2x = 16$
3. Divide both sides by 2.	$x = 8$
4. Interpret: the question may want the sister.	Emma 8, sister $8 + 4 = 12$ ✓

Emma is 8 (and her sister 12) — solving gives $x = 8$; interpreting turns x into each age. Check: $8 + 12 = 20$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (writing expressions, solving two-step equations, and interpreting an answer in context), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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